

16th oct
2020

TOPIC

LOGISTICS

REGRESSION

Logistic Regression

It's a ~~new~~ one more Algorithm that today we are going to ~~for~~ discuss.

This is a 'classification' problem statement.
It is ~~use~~ useful to solve the 'classification problem'.

Let's say,

A person wants to clear a ~~IIT~~ (IIT) JEE. Exam.

If a student study for a 2 hrs daily, will he able to clear the IIT?

Let's me create a features.

<u>Study</u>	<u>play hrs</u>	<u>output (pass/fail)</u>
1	8	Fail
2	7	Fail
3	7	Fail
6	3	PASS
1	4 + 3 hrs	PASS

So, A person is study for 1 or 2 hrs every day & play for 8 to 7 hrs, they will obviously fail.

* If a person is 'Study' for 6 to 7 hrs everyday & play for 3 only, there is a high possibility that person will clear the Exam & pass.

* This all information, I am only ~~just telling~~ telling.

whether a person will 'pass or Fail'?

~~Similar~~

* Similarly, Can we make the application, do the same thing. Yes, there will be some outliers.

* The 'outliers' will be that person will 'Study' for less hours playing for more hours. then only he will clear Exam of ~~IT~~ (IIT)

that specific way, the person will be kind of outliers.

* what is an outliers?

This should have been the result, but this was completely exceptional.

There are some people, who are capable enough study for few hours, but clear every exams.

* There is one person, who will be 'study' for '1 hour' ~~ph~~ & playing for '4 to 5 hours.'

Eating, drinking doing ~~ex~~ extra activities for '3 hrs' still, he/she is able to clear the exams.

why because he is an ~~ex~~ exceptional, he is an ~~out~~ 'outliers' completely.

* There 2 person.

person 1: He always wake up early, bath puja & study & all. Before exam ~~play~~ play.

person 2: He did not care much about

when to study & ~~when~~ we did not even
bath every day. playing games late night.
~~But~~ just say chill we will do
it later on. Study hours also less.

But at result they are able to clear
the exam,

They are rare people, exceptional
people, they are ~~out~~ 'outliers'.

Now, As a human being, we are able to predict
this.

Can we make 'machines' also predict
all this, based on this data.

* How we human says.

like : He is 'Study' for 1 hours,
play for 8 hrs & 'Fail'.

how we are saying this because
our brain is saying this, we are
experienced.

By seeing past or ~~real~~ relatives or our own circle friends.

we say, if he did not study obviously, he will going fail the exam.

by that specific information, we are able to ~~tell~~ tell the 'output'.

Question 'Regression problem'

Can we make a 'machine learning' model do the same exact thing?

Similarly, with the help of 'Regression' problem

Statement

what happened in the 'Regression' problem statement.

* Let's say, we are builder in ~~Bangla~~ Bangalore

Some body ask, Hey what is average price of the house, probably in a Electronic city? one area of Bangalore?

If we are builder, we have some experience. we will be able to say the price of
net?

Yes, obviously we will be able to say the price of the house.

* How that specific price is coming?

Some past information from some past data.

we are trying to automate the entire thing through a model.

But that is 'Regression problem'.

* But right now in 'logistic' problem, we are trying to solve a 'Classification problem'.

① * whether this person will going to pass the exam or not?

② * whether this person ~~to~~ buy a product or not?

Let's discuss this,

Let's say, I have a feature.

Let's take a dataset:

I will take here only 2 features.

① Independent feature & ② Dependent feature

Study Hours

Study Hours
(Independent feature)

output feature

PASS/FAIL

(Dependent feature)

2

FAIL

3

FAIL

4

FAIL

5

FAIL

6

~~FAIL~~ PASS

7

PASS

8

PASS

9

FAIL

① Let's say,

A person study for a exam
'2 hours' every day, for the exam like
'UPSC'
The final output they will 'Fail'
the exam.

② Let's say, for 3, 4, & 5 hours

If a person study for a exam
'3 hours' every day, ~~for~~ then the final
output they ~~to~~ will 'Fail' the exam.

③ Let's say,

~~for the~~
If a person study for a exam
'6 hours' every day then the final output,
he will high probability that
'PASS' the exam.

④ Similarly,

If a person study for a exam

for 7, 8, hours, then they will 'PASS' the exam.

⑤ But,

there are person, who study for a 9 hours every day, but still they are not able to do it, & they 'FAIL' the exam these people are called the 'outliers'.

• But, in 'UPSC', there are hardly some (90) percentage of people get selected, But if I talked ~~about the same~~ with respect to the productivity they are study.

Spending this many hours.

They may most of the time, they may pass.

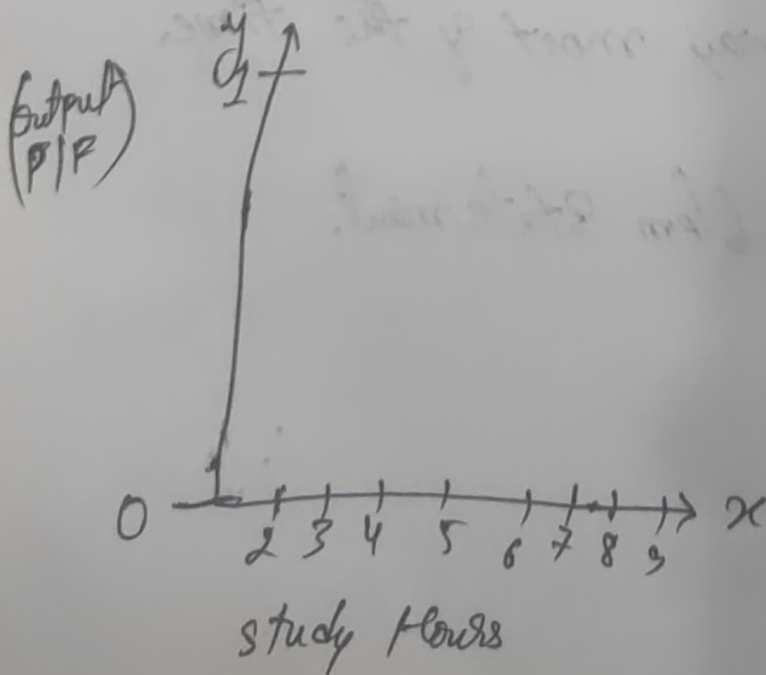
• This is a problem statement.

* Question 5

Can we solve this problem using Regression?

Study Hours	output (pass/fail)
2	F
3	F
4	F
5	F
6	P
7	P
8	P
9	F

I have my dataset, x -axis = No. of Study Hours.
 y -axis = No. output - ~~P~~ F/P.



Now, how many outputs I have in 'y'-axis?

Either it's a ① or either it's a zero(0).

~~Before~~ Because,

FAIL & PASS

①

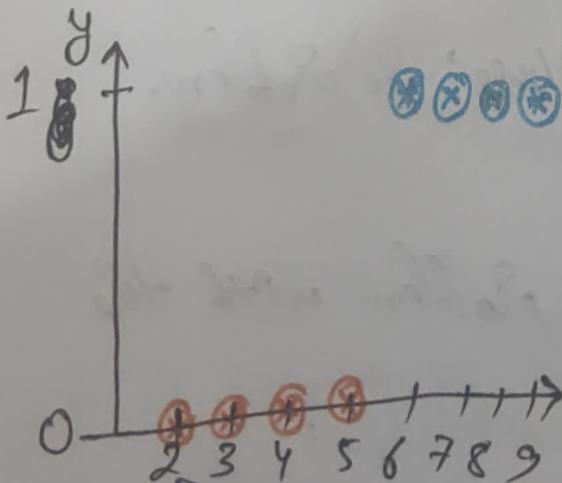
①

If I do this, if I convert this into a 'Categorical Encoding' which is a (OHE) One Hot Encoding.

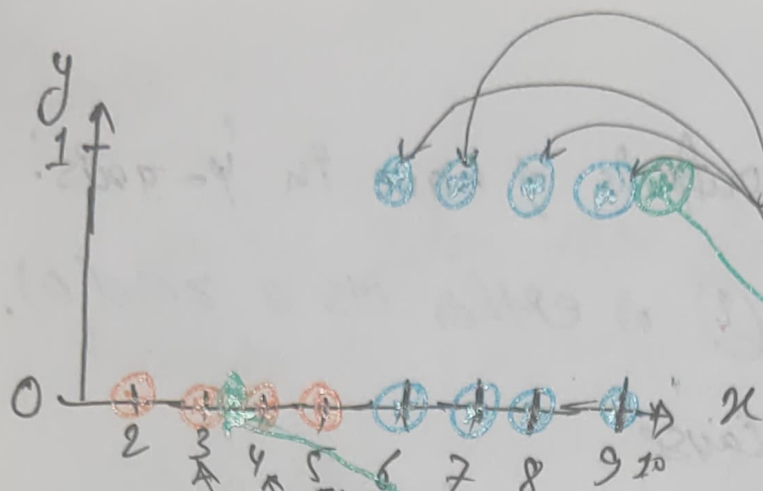
I can make the PASS = 1

FAIL = 0

Now, if I start plotting this in graph.



* ① In ② it is saying 'FAIL' then basically it comes over here.

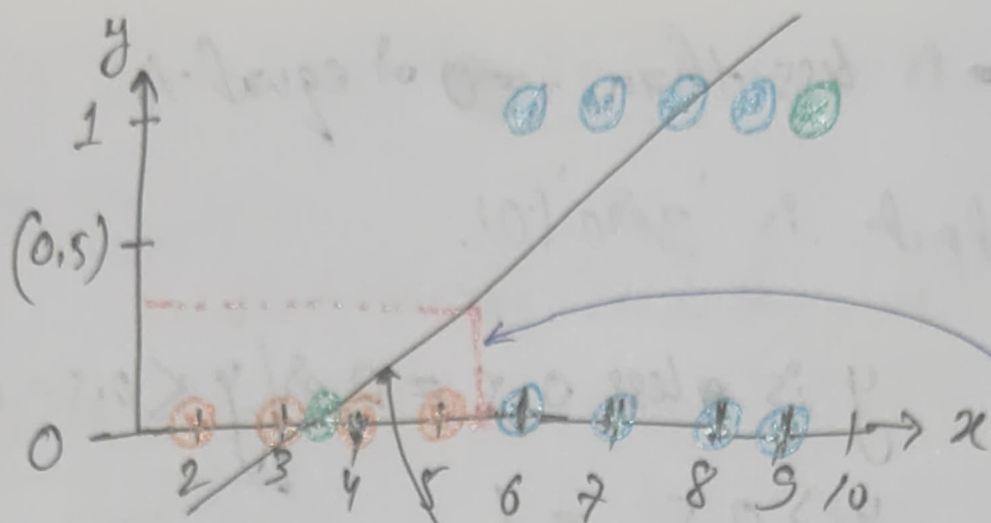


② Now, 3, 4, 5, it is 'FAIL' so, it is coming over here

③ Now, 6, 7, 8, 9 these are 'PASS', so, I will start plotting this over here.

* There ~~are~~ might be some 'outliers' over here.
There might be 'outliers' no problem.

* Now, in order to solve this problem with the help of 'regression'.



* If this is 0 & 1, can I write this model point as a (0.5)

what, we will do is that, we try to create a 'best fit line'

let's say, this is my 'best fit line'

with the help of ~~regression~~ 'regression', we try to create a 'best fit line'.

Through this 'best fit line', we can also make an assumption that, it with respect to this new 'Test Data'.

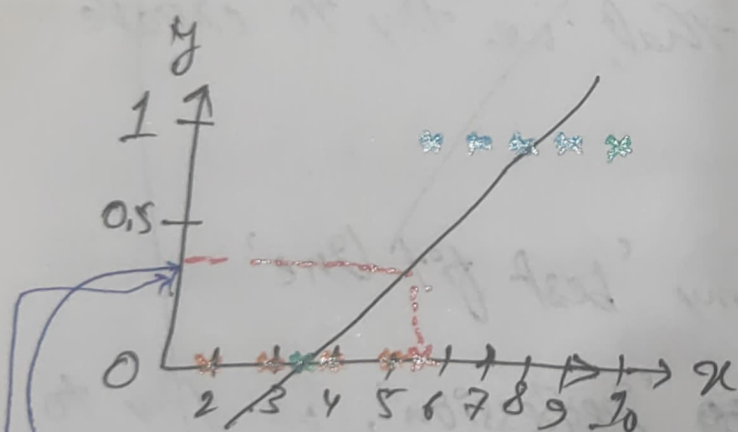
I will just try to plot over here & I will just see ~~the~~ that, where it is coming.

If it is coming 'y' is less than or equal to less than 0.5.

'y' is less than ~~...~~ or equal to
then output is 'zero' (0).

$$y \text{ is less } 0.5 = 0 \text{ or } \boxed{y < 0.5 = 0}$$
$$y > 0.5 = 1$$

If my 'y' is greater than the $\boxed{0.5}$, then
output is $\boxed{1}$.



For this particular point, I will go ahead &
plot it.

what is coming over here.

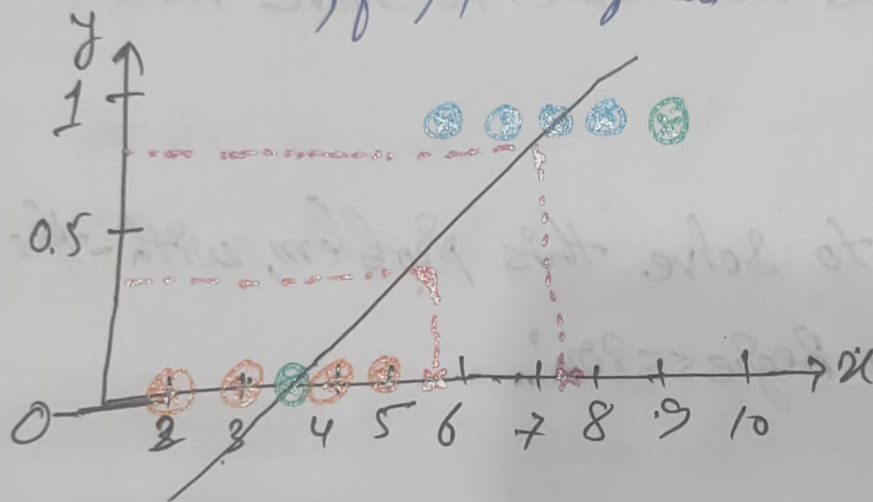
Is this less than $\boxed{0.5}$?

Can I say in this point $\boxed{0.5}$ is my ~~threshold~~
'threshold'.

If y is less than $\boxed{0.5}$, I will consider this as an 'zero'.

that means, it belongs the category, zero.

* Let's say, if y is greater than (0.5) .



Let's say, one point over here, if I try to plot this based on the 'best fit line'.

Here, we can see, I am getting greater than $\boxed{0.5}$.

If I am getting greater than $\boxed{0.5}$

Can I ~~also~~ assume, this, it will be $\boxed{1}$

$$\boxed{y > 0.5 = 1}$$

that is what we are trying to solve it.

* Question:-

If this is being able to solve why we need "Logistic Regression"?

Because with the help of 'Linear Regression', we are able to solve this problem.

Are we able to solve this problem, with the help of 'Linear Regression'?

Note:-

For any new value, I just try to plot this in 'best fit line', I get the output.

If the output is greater than the $\boxed{0.5}$, I am actually being able to say that, OK its belong to $\boxed{1}$ Category.

$$\boxed{y > 0.5 = 1}$$

If it is less than $\boxed{y < 0.5 = 0}$ $\boxed{0.5}$ then it belongs to zero (0) Category.

* So, if I am able to solve with the
'Linear Regression'.
why do I have to go with the
~~Logit~~ 'Logistic Regression'?

Note :-

~~By~~ By default we are saying that
 $[0.5]$ is a 'threshold'

~~* Why Linear Regression will not work for
Classification?~~

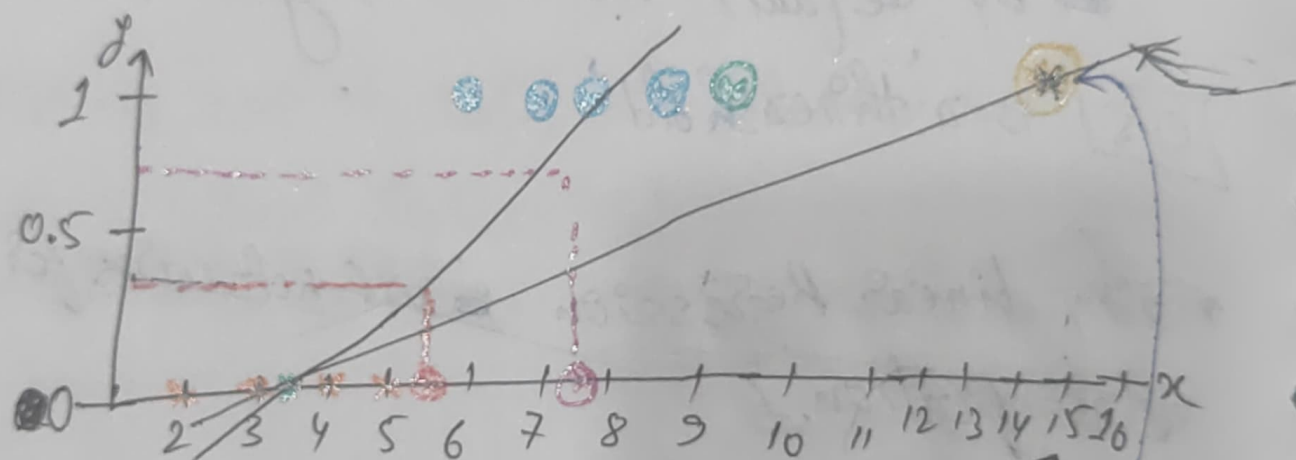
Question:-

why linear Regression will not work for classification?

there is a region for that.

~~Let me~~

if I extend this line 'x-axis'.



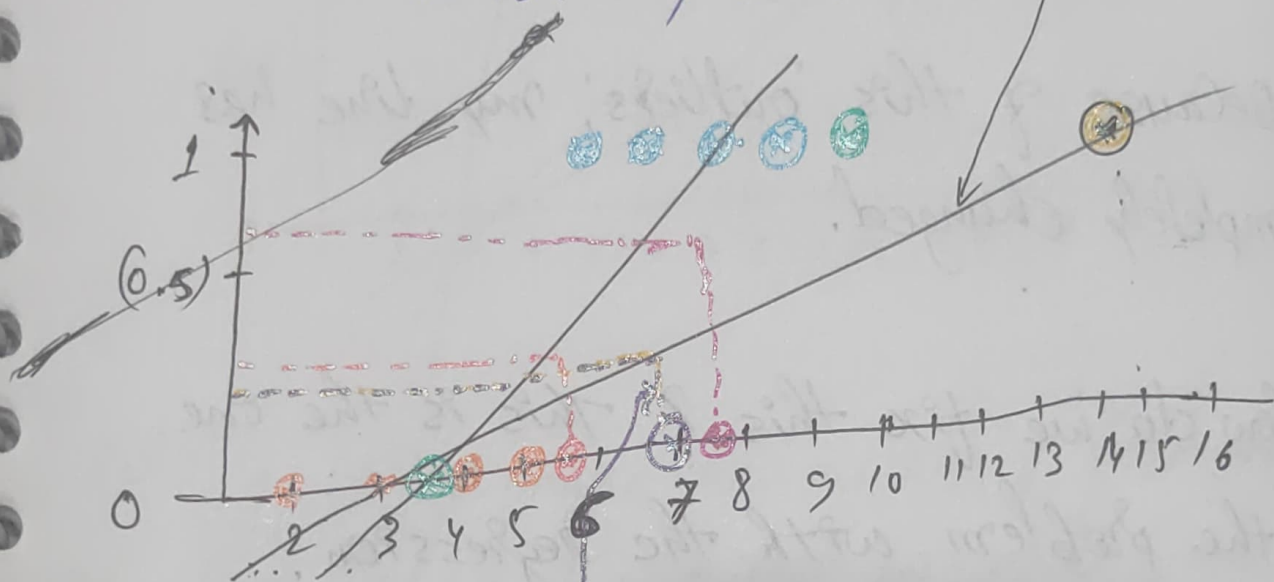
if a person study for 15 hours, what tentative ~~data~~ output based on the dataset we may get?

obviously the person get 'PASS'.

most of the time ~~in~~ the datapoint will come over here.

Now, this 'outlier' coming over here. Don't we think this 'best fit line' will again change.

Because of this outlier, I may get another 'best fit line' which may look like this.



what is the difference, it happening.

Let's say,

I go ahead & compute it for $[7]$.

For the $[7]$ I try to plot it, Am I getting the less than $[0.5]$ Because of this change?

If I am getting less than $[0.5]$ that means my model is predicting 'wrong'.

Even though the person is

studying for $[7 \text{ hours}]$

It is showing that, probably a probability is less than $[0.5]$. It mean person is $[FAIL]$.

* why we can't use the 'Linear Regression' for classification problem?

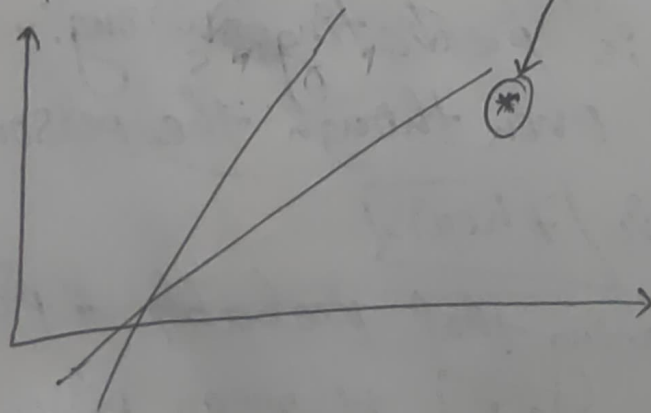
Because of this 'outliers', my line has completely changed.

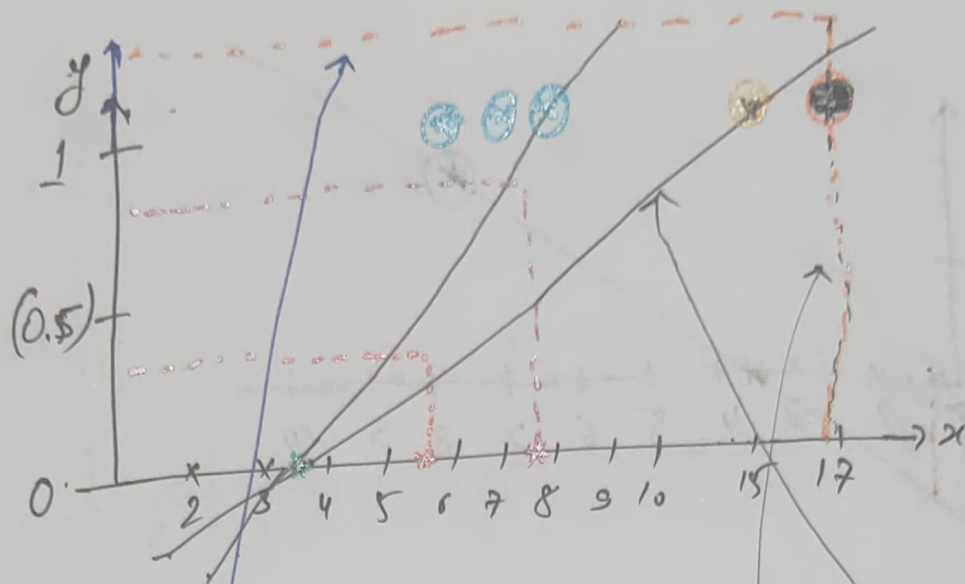
* How do we fix this? This is the one of the problem with the regression.

Because of the some of the 'outliers' my line will completely changed, because of this my result will completely different.

* Note:-

we can't remove the 'outliers' every time, this is super important.





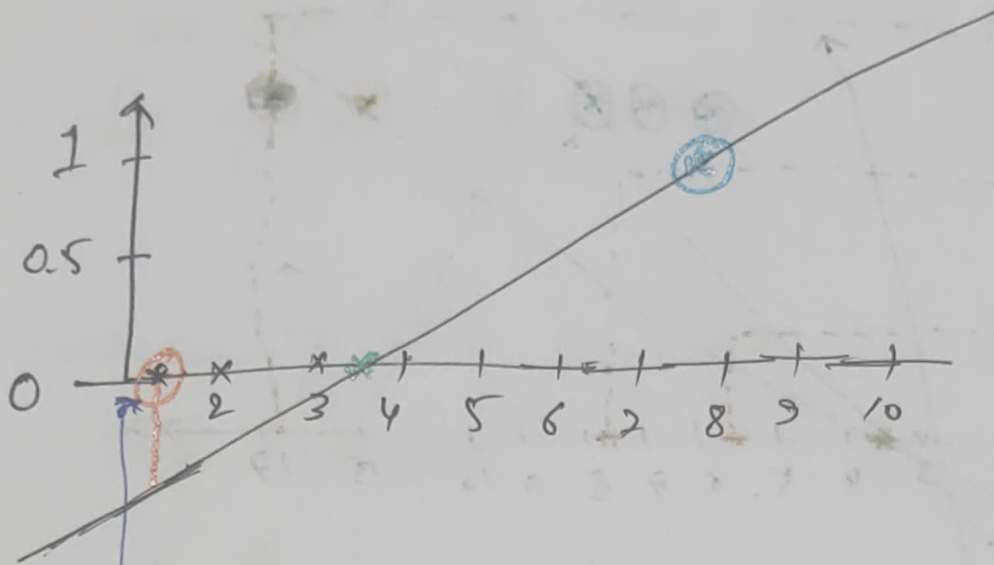
Let's, if I will make this as an 'best fit line'.
 Q: what will be the output for point like
 '17 hours' if I will plot over here
 will it greater than 1?

Yes, it will be greater than 1. That is true.
 I have only these 2 condition.

$$\begin{cases} y \leq 0.5 = 0 \\ y > 0.5 = 1 \end{cases}$$

Now, I am not able to handle this current
 condition, because it is a ~~greater~~ 'greater than 1'
 it crossed the line of boundary.

Similarly,
 if we look into the 0 (zero) below
 part not the upper part.



Let's say, ~~we~~

~~we~~ we have a some value over here. what will be the output, with respect to these? ~~it will be less~~

It will be less than the zero.

* If greater than 1 & less than '0' can be handle?

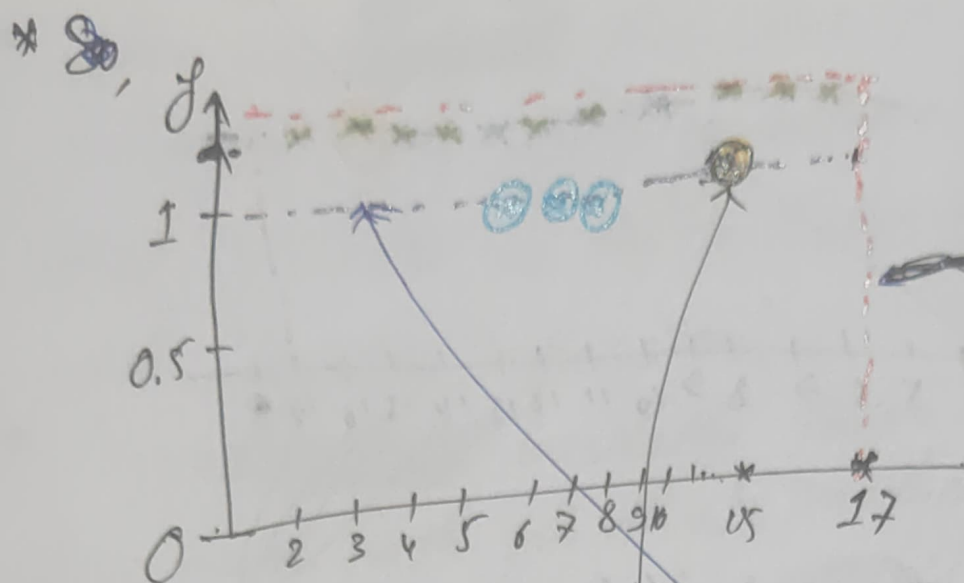
$$\boxed{>1 \& <0}$$

$$\boxed{>1 \& <0}$$

~~But~~ No. right,

But we can handle it. anything greater than $\boxed{1}$, it still be $\boxed{1}$.

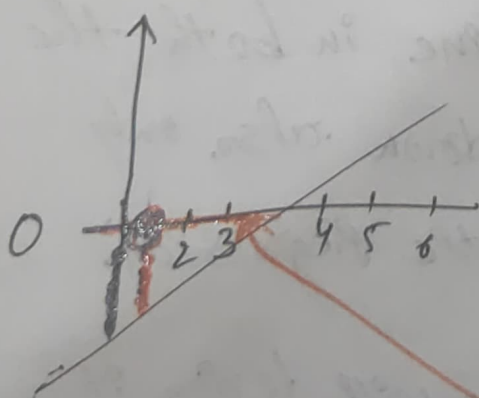
we can handle than.



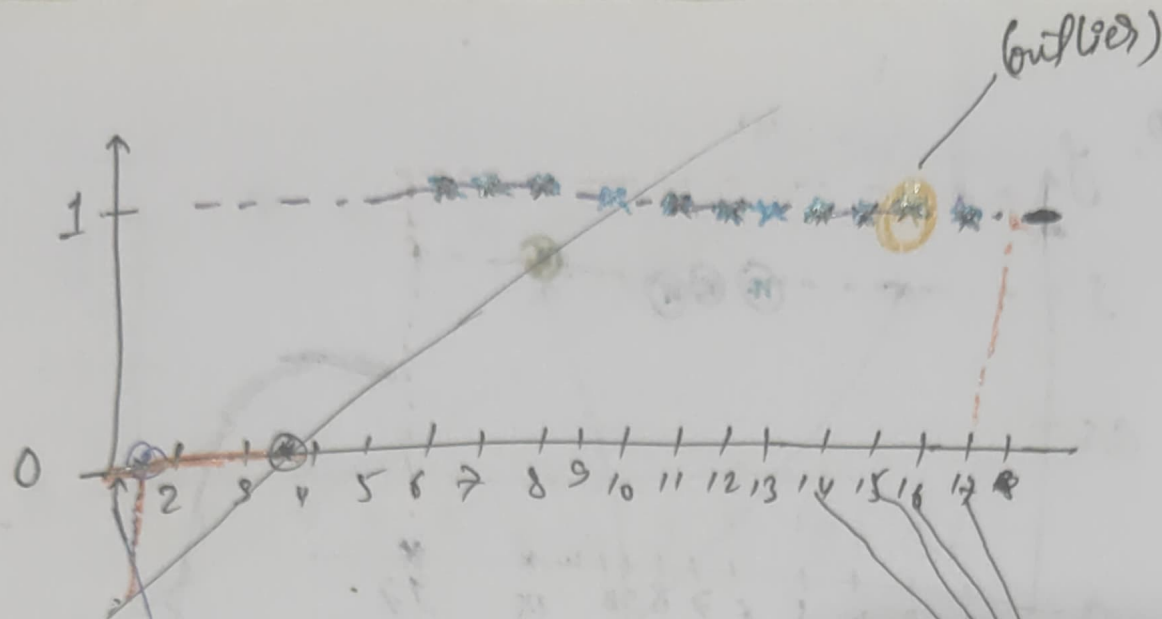
* our main main should be that, even though there is an ~~outlier~~ 'outlier'.

focus on
↑ it should not change this line, instead making the line straight.

* Below down, also.



Here also, it ~~should~~ instead of changing the line, it should focus on changing the line straight.



with respect to the every data point, over here & ~~here~~ here.

I will try to plot it, it should 'squared' it over here.

It should not cross or go above this line.

* This is only possible because of the 'Logistic Regression'.

To squared the line, in both the side above one & below down also, only possible because of 'Logistic Regression'.

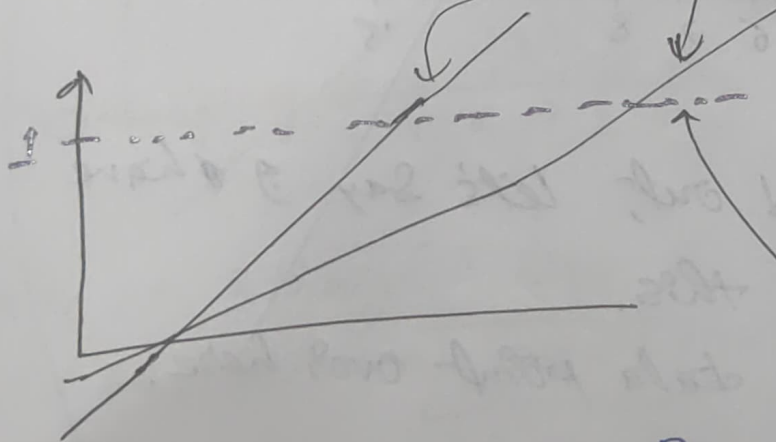
mathematic concept, we will learn it.

* So, because of this 'outlier', we do not change the line. Instead we squared this line mean go straight. Don't change the ~~line~~ ^{line}.

* when we ~~stretch~~ 'squash' this line or make straight line, everything between $[0 \text{ to } 1]$ that is where we try to solve the problem with 'logistic regression'.

* 'squash' ~~line~~ basically mean cut the line make path for that another data point. So, that output only be ~~not~~ between $[0 \text{ to } 1]$ not crossed the line.

* let's say,
This is our 'best fit line', but after



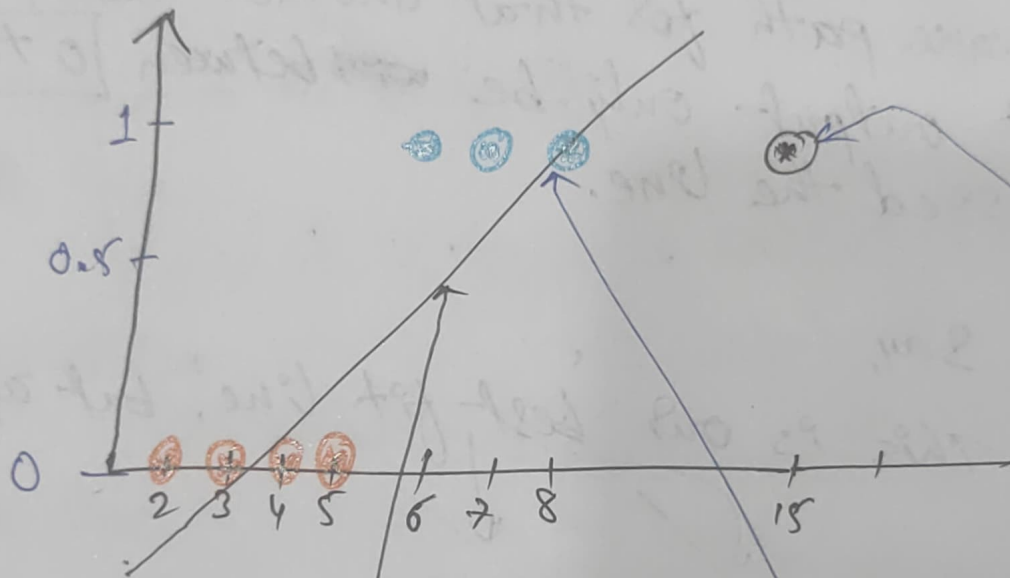
this analysis, cut it from ~~the~~ here, don't make ~~the~~ 'best fit line' longer.

* Note :-

Revision

what we learn, better understand it short. * revision below.

what we trying to do over here is that,



we are trying find out, let's say I have a data point like this.

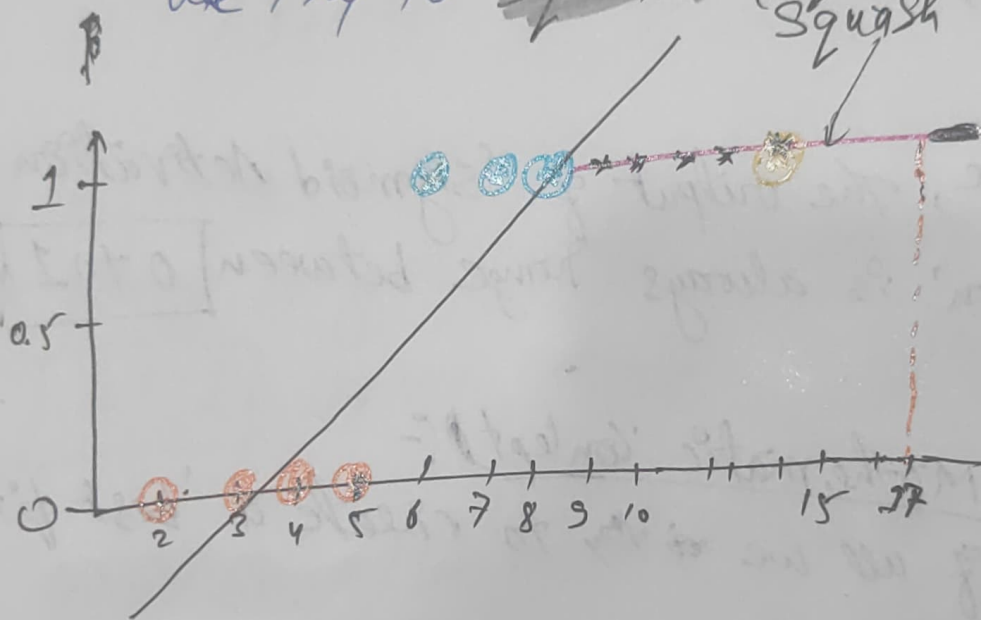
this data point over here.

* our condition was why we should not used the 'Linear Regression' to solve this Classification problem.

Because, there one 'outlier' comes, the this 'best fit line' completely changed.

we don't want this, so instead of that,
 Even though 'outliers' is there, we have
 to create a 'best fit line'.

To make this 'best fit line' get less
 impacted from this, we try to do is that
 we try to ~~squashed~~ 'squash' this.



make it a straight line. So that all my output
 will be between 0 to 1.

* This squashed is done by important function
 is called as 'Sigmoid Activation' function.

* 1st step, this is my 'best fit line'

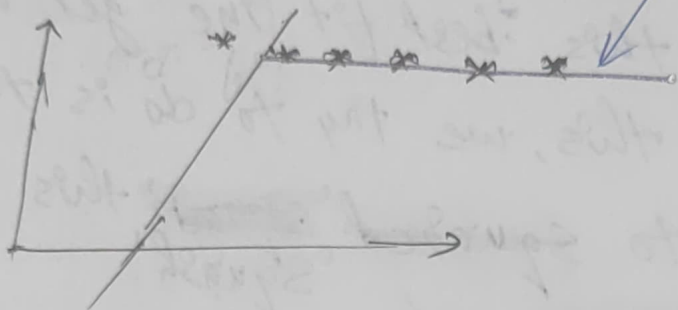
we calculate it,
$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

we use same equation, but what

we do, we apply the → 'Sigmoid Activation Function'

once we applied 'Sigmoid Activation Function'.

our 'output' this 'Sigmoid' will happen.



Because, the output of 'Sigmoid Activation function' is always range between $[0 \text{ to } 1]$.

* Mathematic Concept:-

1st of all we try to create a 'best fit line'.

which is (1) $h_0(x) = \theta_0 + \theta_1 x$

Lets put variable (z)

$$z = h_0(x) = \theta_0 + \theta_1 x$$

(2) 2nd step. we applied a 'Sigmoid Activation function'.

which is

$$\text{Sigmoid function} = \left(\frac{1}{1 + e^{-z}} \right)$$

what is 'z', Here I have assigned.

$$z = 0_0 + 0_1 x$$

whenever, we applied this ~~for~~ ^{equation.}

$$\frac{1}{1+e^{-z}}$$

* Let 'z' value be anything. (+ve) or (-ve) anything, we always going to get the value of between 0 to 1. If we calculate in Calculator also

~~$\frac{1}{1+e^{-20}}$~~ type inside Calculator

①

$$1 \div (1 + e^{-20})$$

$$= 0.999999 \dots$$

we are getting below than 1 value.

② $1 \div (1 + e^{2000})$

$$= 0$$

* Notes

what this 'sigmoid Activation function' is doing?

Sigmoid function making sure that, whenever we applied, on top of this $[z]$ it is always going to give the output between

~~0 to 1~~ $[0 \text{ to } 1]$.

This is what we want between

$[0 \text{ to } 1]$.

Through this basically we are able to solve the 'classic problem'.

Question :-

what is 'Linear Regression' Cost Function?

With the help of 'Linear Regression', we can even derive, CNN, RNN whatever things we want, that is why this is super important.

$$J(\theta_0, \theta_1) = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

This is my equation.

what is $h_{\theta}(x^{(i)})$ over here.

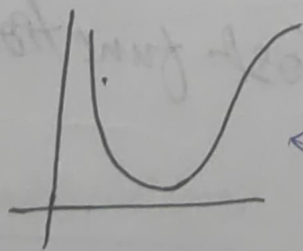
$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

$$y^{(i)}$$

this is my cost function, what is so important about this.

this 'Cost function' is nothing but, it is (MSE). (Mean Squared Error).

what is superb about the MSE?



we get an 'Gradient descent' like this.

If we are getting 'Gradient descent',
Can I say this create a 'Convex function'?

In 'Convex function', we get a one 'Global minima', it is a 'Quadratic equation'.

~~what is logistic regression~~

Question: 2

what is ~~Loges~~ 'Logistic Regression'
Cost function?

In the logistic regression @ 1st step is
to create an 'Best fit line'.

(b) Then we have to 'squashing' squash,
with the help of 'sigmoid function'.

* 'Sigmoid Activation function', again comes
in 'Deep learning'.

Can I write this 'cost function' :-

~~J(θ₀, θ₁)~~

$$J(\theta_0, \theta_1) = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

$$h_{\theta}(x^{(i)}) = \sigma(\theta_0 + \theta_1 x)$$

↑
Sigmoid Activation.

~~h₀~~
* $h_0 x^{(1)}$ how do I write?

I will use σ notation, 1st of all, I need to apply a sigmoid Activation function (σ) on top of this. this equation $(\theta_0 + \theta_1 x)$ or best-fit line.

(σ) - This sigma is not the standard deviation, this is a "sigmoid Activation function" function-notation.

* Let's use another better notation for this.

Let's, I am using
$$z = \theta_0 + \theta_1 x$$

$$= \sigma(z)$$

~~h₀~~
It will become (σ) sigma on top of the (z).

~~(σ)~~ Sigmoid (σ) is nothing but

$$\sigma = \frac{1}{1 + e^{-z}}$$

~~$$\frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$~~

$$h_0 x = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}} \quad \leftarrow \begin{array}{l} \text{replace } (z) \text{ over} \\ \text{here} \end{array}$$

So, this is how my final outcome equation looks like,

$$h_0 x^{(i)} = \sigma(\theta_0 + \theta_1 x)$$

$$= \sigma(z)$$

$$= \frac{1}{1 + e^{-z}}$$

$$h_0 x^{(i)} = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}} \quad \checkmark$$

~~Here,~~ ~~I given notation~~

Question: How z came?

we have assigned it, let $\boxed{z = \theta_0 + \theta_1 x}$

I am giving notation to this $\boxed{\theta_0 + \theta_1 x}$,

write $\boxed{z}$ to write more.

~~let's un~~

Note: let's understand 'logistic regression' again. ~~repeat~~

① create ~~best~~ best fit line 1st.

② Squashing with Sigmoid Activation function.

~~what will be~~

$$J(\theta_0; \theta_1) = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

~~$h_{\theta}x = \theta_0 + \theta_1 x$~~

$$= h_{\theta}x = \text{sig}(\theta_0 + \theta_1 x)$$

$$= h_{\theta}x = \text{sig}(z) \longrightarrow (z = \theta_0 + \theta_1 x)$$

$$= h_{\theta}x = \frac{1}{1 + e^{-z}}$$

$$= h_{\theta}x = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

output

0 to 1

what will be my $h_{\theta}x^{(i)}$ or $h_{\theta}x$?

It will be same nothing $[\theta_0 + \theta_1 x]$

But, in our 'logistic regression', what we add?

we add the 'sigmoid activation function'.

If I am adding sigmoid function,
we can't write like this one.

$$\boxed{\cancel{h_0 x} + \theta_0 + \theta_1 x}$$

we have to add the sigmoid function
on top of it.

let's ~~denote~~ denote as (sig).

$$h_0 x = \text{Sig}(\theta_0 + \theta_1 x)$$

Now, I will apply this sigmoid function (sig)
on top of the 'best fit line'.

Best fit line given by ~~to~~ $(\theta_0 + \theta_1 x)$.

* Now, Let assume,

~~Let assume~~ $\boxed{z = \theta_0 + \theta_1 x}$ So that we don't

have to write it again & again.

we can say ~~variable (z)~~

Let's take variable as (z)
Instead of ~~we~~ writing $(\theta_0 + \theta_1 x)$.

So,

$$h_{\theta}(x) = \text{sig}(z)$$

* How sigmoid notation given by:

$$\frac{1}{1 + e^{-z}}$$

So, then finally my $h_{\theta}(x)$ will be:

$$h_{\theta}(x) = \frac{1}{1 + e^{-z}}$$

Now, I will replace this (z) over here

So my final

$$h_{\theta}(x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

because $[z]$ is nothing but, $[z = \theta_0 + \theta_1 x]$

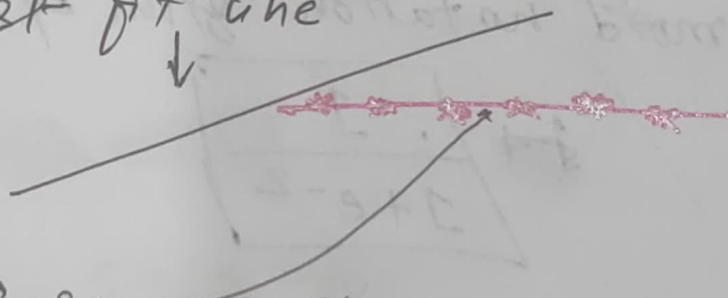
This is my final notation.

By this equation

$$h(x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

I am able to do 2 steps.

* create best fit line



* & 'squash' it out it.

Because the output will be always

~~the~~ between $[0, 1]$

~~threshold~~

* Threshold

$$(1) \leq 0.5 = 0$$

$$(2) > 0.5 = 1$$

* If it is less than $[0.5]$ my output will be '0' [Fail]

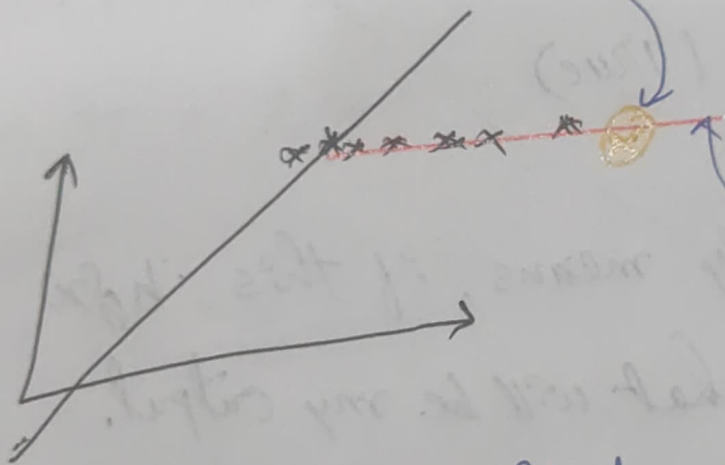
* If it is greater than (0.5) my output will be $[1]$ [Pass]

* Question :-

Do we add to sigmoid function,
if the 'outliers' is there?

Yes, it is solving that problem
as well.

If the outliers, present over here also,



Since, we are 'squashing' over here, it will
not create a very big impact.

52:28

* Threshold :-

$$\sigma \leq 0.5$$

$$* \quad h_0(x) = \frac{1}{1 + e^{-(0.0 - 0.9x)}}$$

Threshold :

$$(1) \leq 0.5 = 0 \text{ (Fail)}$$

$$(2) > 0.5 = 1 \text{ (True)}$$

0.75

* That basically means, if this $h_0(x)$ is 0.75 then what will be my output.

~~It is~~ The output will 1

* This particular output ~~gives~~ 0.35 what should be the output?

The output would be (0) zero.

* ~~the output~~ ^{It} gives 0.25 what should be the output?

The output would be (0) zero.

$$\textcircled{1} \leq 0.5 = 0 \text{ [Fail]}$$

$$\textcircled{2} \geq 0.5 = 1 \text{ [True]}$$

$$\textcircled{3} 0.75 = 1 \text{ [True]}$$

$$\textcircled{4} 0.35 = 0 \text{ [Fail]}$$

$$\textcircled{5} 0.25 = 0 \text{ [Fail]}$$

$$\textcircled{6} 0.95 = 1 \text{ [True]}$$

~~$\textcircled{7} 0.5$~~

$$\textcircled{7} 0.59 = 1 \text{ [True]}$$

This will be my outputs.

* Non-Convex Function :-

Note :-

But there is a one big issue with this equation is that, whatever we have learn till now.

Cost Function will lead to some kind of 'convex function' only or 'non-convex function'.

But, when we apply this ~~to~~

$h_0(x)$ with this value

$$\frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

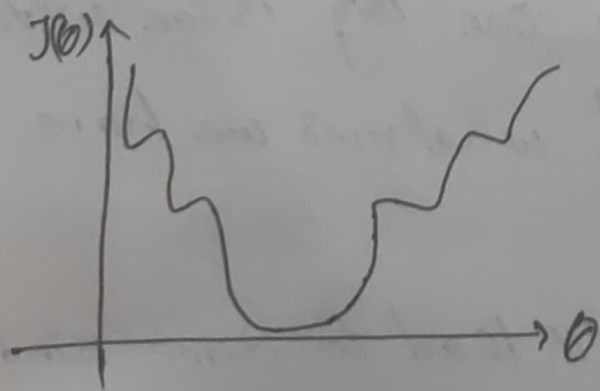
then when we use this 'cost function.'
this will create a Non-Convex function.

***Question:**

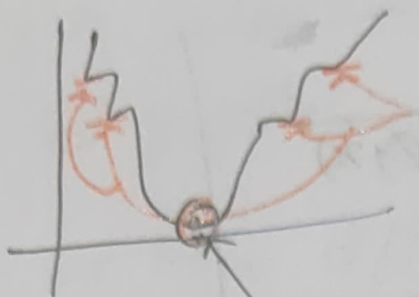
difference between Convex Function &
Non-Convex Function?

54:24

① Non-Convex Function :-



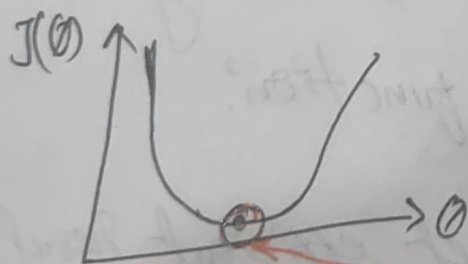
Here we will get 'Non-Convex' like this.
function



In ~~convex~~ function, there is a lot of 'local minima',
 there is high chance that 'local minima' get stuck. & we have a one 'Global minima'.

Q2) Convex Function :-

Convex function specially, we use for linear regression, we use for quadratic equation.

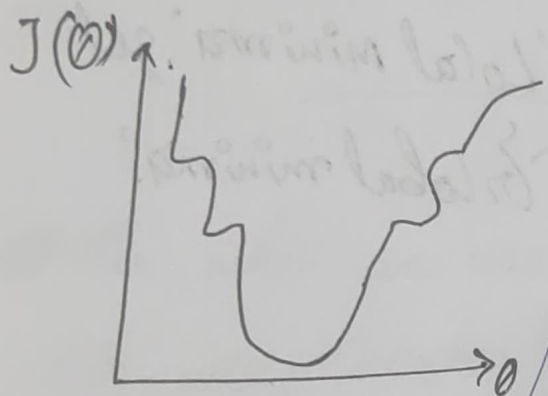


Here we use to get one 'Global minima'.

* Note :-

But ~~now~~ right now whatever things we are using its leads to this 'Non Convex function'.

$$h_0(x) = \frac{1}{1 + e^{-(\theta_0 - \theta_1 x)}}$$



that is the issue, when I try to plot $J(\theta)$ with this, this was actually leading to this kind of 'Non-convex function'.

we are not able to get efficient result in the backend.

Fix problem statement:-

* Now, how do we fix this problem statement?

Simple way to fix particular problem is to simply change 'cost function'.

now we have different 'cost function' all together.

we will be not using this 'cost function'.

$$J(\theta_0, \theta_1) = \frac{1}{m} \sum_{i=1}^m (h_0(x)^{(i)} - y^{(i)})^2$$

when $h_0(x)$ is this specific value.

$$h_0(x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

we will using different 'cost function', that is called 'Log Loss Cost function'.

* Log Loss Cost function :-

* this is all through research, end of the day this particular model is not performing well. there maybe issue some issue.

the deep dive into it, they start plotting it, they saw that there is a lot of 'Local minima' was actually coming.

'Local minima' actually comes, updation

* ~~Log Loss cost f~~

If co-efficient will get stuck over there. That is the region, we don't use any model, instead we focus on 'Log loss

for cost function'.

So, we we don't go with ~~that~~ this 'cost function'

$$J(\theta_0, \theta_1) = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

* Log Loss Cost Function:-

So, the cost function is with respect to this, is given by this notation.

$$\text{Cost}(h_{\theta}(x^{(i)}), y^{(i)})$$

what is $h_{\theta}(x^{(i)})$ given by this particular equation $\rightarrow$

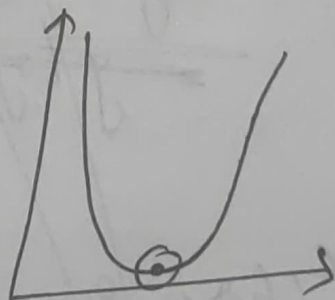
$$h_{\theta}(x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

This cost function is nothing but

$$\text{cost}(h_0(x^{(i)}; y^{(i)})) = \begin{cases} -\log(h_0(x)) & \text{if } y=1 \\ -\log(1-h_0(x)) & \text{if } y=0 \end{cases}$$

* what is $h_0(x)$ is $\frac{1}{1+e^{-(\theta_0+\theta_1 x)}}$

when we use this equation, at the end of the day, we will going to see this kind of 'Gradient Descent' to find out the 'best fit line'.



If we expand this 'log loss cost function' then, it will be written something like this.

$$\text{Loss} = \text{cost}(h_0(x^{(i)}; y^{(i)})) = -y \log(h_0(x)) + (1-y) \log(1-h_0(x))$$

Formula :-

$$\text{Loss} = -[y \times \log(h_0(x)) + (1-y) \times \log(1-h_0(x))]$$

means:-

y = actual label (either 0 or 1)

h_0 = model's predicted probability (output of sigmoid)

$\log$ = natural logarithm

(1) $y = 1$

$$\text{Loss} = -[1 \times \log(h_0(x)) + (1-1) \times \log(1-h_0(x))]$$

$$= -[\log(h_0(x)) + 0 \times \log(1-h_0(x))]$$

$$= -[\log(h_0(x)) + 0]$$

$$\boxed{\text{Loss} = -\log(h_0(x))}$$

(2) $y = 0$

If the $y = 0$ then,

$$\text{Loss} = -[0 \times \log(h_0(x)) + (1-0) \times \log(1-h_0(x))]$$

$$\text{Loss} = -[0 + 1 \times \log(1-h_0(x))]$$

$$\text{Loss} = -\log(1 - h_0(x))$$

answer

$$\boxed{\text{Loss} = -\log(1 - h_0(x))}$$

* So,

If $\boxed{y = 1}$ then we will get the final

$$\boxed{\text{Loss} = -\log(h_0(x))}$$

*

If $\boxed{y = 0}$ then we will get the final.

$$\boxed{\text{Loss} = -\log(1 - h_0(x))}$$

So

Basically,

~~This is the 'Log Loss Cost function'~~

$$\text{Cost}(\text{model}, y) = J$$

Basically, this is the 'log loss cost function'

$$\text{Cost}(h_0(x)^{(i)}, y^{(i)}) = -[y \times \log(h_0(x)) + (1-y) \times \log(1-h_0(x))]$$

we are basically using, this 'cost function'
we will create a 'convex function'.

* what is 'y'?

Can we say my 'y' is the truth value
Because, here we will be able to see that,

In the case of 'Linear Regression' we saw
that 'cost function' basically we use.

Now,

If we see over here my 'cost function'
will look like this.

* why ~~we~~ this 'cost function', we will not
gonna use?

$$J(\theta_0, \theta_1) = \frac{1}{m} \sum_{i=1}^m (h_0(x)^{(i)} - y^{(i)})^2$$

Because, it is a 'Non-Convex' function.

Now this 'cost function' below.

$$\text{Cost}(h_0(x^{(i)}, y^{(i)})) = -[y \times \log(h_0(x)) + (1-y) \times \log(1-h_0(x))]$$

The researcher started researching for this kind of this what can be definitely used.

So, finally they came up with this

'Log Loss Cost function'.

which many people call with different different name.

- ~~(A) Likelihood~~
- (A) Likelihood function
- (B) Log-likelihood.
- (C) Negative log-likelihood.

Through this, we never get a ~~total minima~~.

'Local minima'

that is the region, instead of using that 'loss function'.

we basically used this 'Loss Function'.

* Now, the remaining steps are almost similar,

* what are the steps?

(1) we minimize the 'cost function', this is 'cost function'.

$J(\theta_0, \theta_1)$ by changing (θ_0, θ_1)

* ~~how~~ how do we change it?

in Linear Regression, if we remember,

'Convergence Algorithm'

what we do in 'Convergence Algorithm', we repeat the all the 'Convergence', in this Convergence, what we do.

$J = 0 \text{ \& } 1$

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0; \theta_1)$$

}

In this 'conversation' what we do?

$J = 0 \text{ and } 1$ Considering, I have one intercept
& 1 co-efficient.

I will write ~~$\theta_j := \theta_j$~~ $\theta_j := \theta_j$ - will
updated with (α) learning rate. Divide by

$\left(\frac{\partial}{\partial \theta_j}\right)$ derivative of θ_j & $J(\theta_0; \theta_1)$.

* ~~How to calculate the derivative, we already~~

* How to calculate the derivative (2)?

we already learn in our previous class.

'Linear Regression'.

- All the steps will be similar to calculate this.

Note:

In 'logistic regression', we were trying to 'squash' it.

Here, in this we have discussed why not to use this 'cost function'.

$$J(\theta_0, \theta_1) = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

Instead of this!

we will going to use this 'cost function' one

$$\text{Cost}(h_{\theta}(x^{(i)}), y^{(i)}) = -[y \times \log(h_{\theta}(x)) + (1-y) \times \log(1-h_{\theta}(x))]$$

~~Ques~~

* Note:-

In interview, they mostly asked about 'Log loss function'?

Then, we just have to tell this scenario, based on this scenario, whatever the 'cost' value will be coming based on that value it get updated. (0, 1) then we just have to reduce this 'cost function value'.

$$\checkmark \downarrow \downarrow \text{Cost}(h_0(x), y^{(r)}) = \begin{cases} -\log(h_0(x)) & \text{if } y=1 \\ -\log(1-h_0(x)) & \text{if } y=0 \end{cases}$$

Summary what we learn?

* 1st we discussed, with the help of 'logistic regression',

why we don't use 'regression' problem statement to solve it.

we understood the 2 main reason.

① If there is an 'outliers', entire 'best-fit line' will change. & we don't want this thing to happen.

So, we try to 'squash' that particular 'best fit line'.

So, we don't get ~~extended~~ extend, more than 1, & less than (0) zero.

This 'squash' is done by the 'Sigmoid Activation function'.

when we apply this formula.

$$\text{Sigmoid function} = \frac{1}{1 + e^{-z}} = 0. \text{ to } 1$$

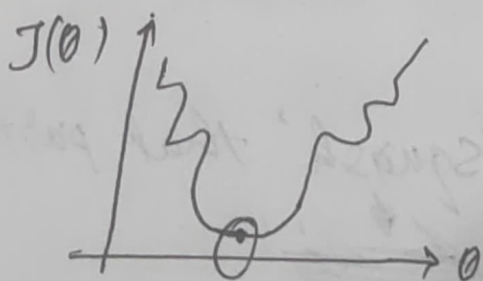
The output is always between to $[0 \text{ to } 1]$.

By 'Squash' we always get the $[0, 1]$,
what we really want to achieve.

Next,

- why we do use same 'cost function'
like the 'Linear Regression'.

Because, it actually creating a
'Non-Convex function'



Since, it is creating a 'Non-Convex function',
~~and~~ what researcher did?

They did a lot of research, they came
up with ~~'Log loss cost function'~~



'Log loss cost function'

public Question *

$$\text{Cost}(h_0(x), y^{(i)}) = \begin{cases} -\log(h_0(x)) & \text{if } y=1 \\ -\log(1-h_0(x)) & \text{if } y=0 \end{cases}$$

$$\text{Cost}(h_0(x), y^{(i)}) = -[y \times \log(h_0(x)) + (1-y) \times \log(1-h_0(x))]$$

which is super important, with the help of this, they started trying to solve this problem.

public Question

why 'y' value more than 2? what happens?

• y value will never be more than 2. It always be 1.

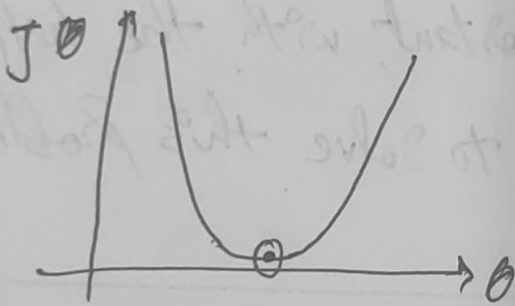
* 'cost function' is a mathematical explanation.

* public Question

~~In logic~~

In logistic regression, we always use non convex function?

No, it is not true. we have to use the convex function



* In logistic regression, we have to find the probability.

So, that is the reason, we took a 'threshold'

of $[0.5]$

How to select the 'threshold'?

There are multiple way, to find it, we will discuss going forward.

By default

By default, I am going to take it as

● $\boxed{\text{threshold} = 0.5}$.

For medical problem, the threshold

● may increase, it may be.

$$\text{0.75} > 1$$

anything $\geq$ greater than $\boxed{0.5}$, so $\boxed{0.75}$ then it will $\boxed{1}$.

'problem statement', something related to ~~vaccat~~ vaccination. threshold will be very very high!

Because, it should be able to give a success rate. threshold, how to use, we will ~~discuss~~ discussed more in detail.

we have to use something call

'ROC - AUC Curve'. later we ~~will~~ gonna discuss.

Performance Metrics

- ① R-Squared
- ② Adjusted R-Squared

With the help of these, we are able to understand the many things, which are important ~~which are not~~ like how much is the accuracy of the model.

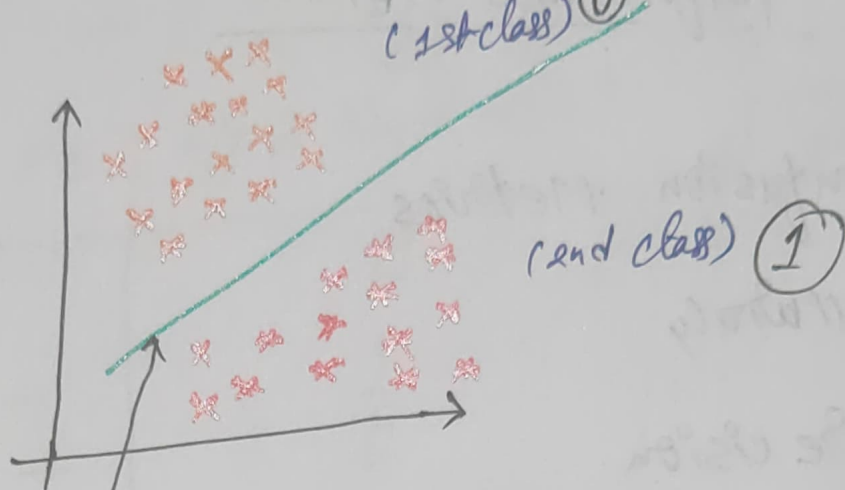
* In ~~classical~~ Classification problem, what do we thing performance matrix should be?

In logistic regression, what we are trying to do.

Let's say,

we have

Let's say,
we have a data points of 2 classes.
(1st class) (0)



with the help of 'Logistic Regression' all I am doing is that, I am trying to calculate the 'best fit line' which will be able to divide this 2 lines.

- * 1st class will be one category (0) zero.
- * 2nd class will be another category (1) one.
- * This is what we are trying to find out.
we are trying to find out the 'Linear best fit line'.

* Now,

How did this model is performing well or not? How do we check it.

Topics Covered

performance metrics

- ① Confusion Metrics
- ② Accuracy
- ③ Precision
- ④ Recall
- ⑤ F-Beta score

Note:-

ROC-AUC will covered next today, next lecture.

- * ROC : (Receiver operating characteristic) curve.
- * AUC : (Area under the curve).

Let's say, we have a 'dataset' (1) Confusion Metrics

F1	F2	(y) o/p or truth output	$\hat{y}$ (model prediction)
-	-	0	
-	-	1	
-	-	0	
-	-	1	
-	-	1	
-	-	0	
-	-	1	

Let's say, in this 'dataset', I have 2 features: F1, F2, with respect to this, I have my output & truth output.

usually, in classic problem, how data points, we have, we have Feature 1 (F1) & Feature 2 (F2) with respect to my 'truth output', I am having

0
1
0
1
1
1
0
1

Can we say, this is my 'Truth output'.

Can we say this is my ('y') variable.

After train our model, it gives a prediction.

Can we write it as $\hat{y}$ or $y(\text{hat})$.

Let's say,

F_1	F_2	dp	y
1	1	0	1
1	1	1	1
"	"	"	"

After my model has been trained, for this particular ~~test record~~ record, it will do the prediction.

Let's say, it has done the prediction 1

Similarly, this way it has done the prediction of all test,

$\hat{y}$
1
1
1
0
1
0
1
0

* $(\hat{y})$: This is my 'model prediction'.
my model has predicted this things.

Let's discuss, 1st topic.

(1) Confusion Metrics : 1:26:26

In Confusion metric, for the 'Binary classification'.

$(\hat{y})$
 1 0 Actual values
 1
 0
 Predicted value
 $(\hat{y})$

Its a 2 cross 2 metrics, why like that ~~for~~
Confusion, because it will gives us a lot of
Confusion.

On the top, we have some 'Actual value'.
It determine by (y) .

on bottom this side, its a 'predicted values.' it determine by $\hat{y}$ or $y(\text{hat})$.

when, it is 0 & 1.

where should, I write it.

	1	0	Actual value (y)
1		1	
0			

Let's say, I am updating 1 over here.
 ↓
 Count of

Next,

(2)

f_1	f_2	(y) O/P	$\hat{y}$
-	-	0	1
-	-	1	1

Now, here in this 2nd scenario, 1 & 1.

	1	0	Actual value (y)
1	1	1	
0			

predicted value ($\hat{y}$)

I can update it over here.
 If my (y) is 1 & my predicted value is also 1.

It is ~~absolutely~~ correct. my model should perform like this only.

Next,

(3)

P_1	P_2	$\frac{0}{P}$	$\frac{1}{P}$
-	-	0	1
-	-	1	1
-	-	0	0

Now, here ~~3rd~~ scenario 0 & 0.

	1	0
1	1	1
0		<u>1</u>

when the 0 & 0 fall, I will update the value. 1 over here

my ~~model~~ correct output is (0) zero,

then my predicted value would be '0' zero,

then the my model is absolutely correct.

Next,

(4)

f_1	f_2	(y) 0/p	$\hat{y}$
-	-	-	1
-	-	1	1
-	-	0	0
-	-	1	1

Now, here 4th scenario $\rightarrow 1 \ 1$

	1	0
1	1	1
0	1	1

my ~~correct~~ correct output is ~~(0/300)~~ $\boxed{1}$ then my,
 'predicted value' would be ~~(0/300)~~ $\boxed{1}$ then
 my model is absolutely correct.

Now,

my final table look like this

~~2~~ 2 cross 2 metrics.

(y)

Actual Record

		1	0
1		3	2
0		1	1
Predicted value ($\hat{y}$)			

Confusion matrix

Now, this table completed.

f_1	F_2	(y) O/P	$\hat{y}$
-	-	0	1
-	-	1	1
-	-	0	0
-	-	1	1
-	-	1	1
-	-	0	1
-	-	1	0

~~20/1~~

* In my dataset, $[0 \& 1]$ How many times,

Came or repeated.

2 times.

	2
1	

* In my data set $[1 \& 0]$ How many times
Came or repeated.

1 times

	1
1	

* In my dataset $[0 \& 0]$ How many times Came
or repeated.

1 times.

1	
1	

* In my dataset $[1 \& 1]$ How many times, it
Came or repeated.

3 times

3	
1	

So, now if we see final table, it would be
look like this

	1	0
1	3	2
0	1	1

* Note:-

• How to we can give notation for this in 2 Cross metrics.

		Actual value (y)	
		1	0
Predicted value ($\hat{y}$)	1	TP	FP
	0	FN	TN

① whenever there, is $[1 \& 1]$, it will give a notation of (TP) True/positive.

It means, it always going to be right. The Actual value is $[1]$ & the predicted value is $[1]$ means my model is doing right prediction.

TP	

② whenever, I see $[0 \& 0]$ this is always also, 'True.'

But we will write it as a True/negative (TN).
Negative, because of zero (0).

	TN

positive = 1

Negative = 0

But it is a correct match, it is being able to predict correctly.

③ whenever, I see 0×1 we give a notation of (F/P) False/positive.

	FP

Can we say, this is a wrong prediction.

④ Similarly, for this one last one also, for 1×0 we give a notation of

(F/N) False/Negative.

FN	

~~what~~
* which are my right prediction?

	1	0
1	TP	FP
0	FN	TN

* These both combined is my right prediction.

* These both combined is my wrong prediction

FP & FN

TP	FP
FN	TN

(2) Accuracy

* Now, How do I calculate my Accuracy?

Can I write my accuracy nothing but

$$\begin{aligned}\text{Accuracy} &= \frac{TP + TN}{TP + FP + FN + TN} \\ &= \frac{3 + 1}{3 + 2 + 1 + 1}\end{aligned}$$

$$= \frac{4}{7}$$

$$= 57\%$$

So, the my accuracy is this much 57% ✓.

(3) precision

Let's say,
I have a 'dataset' & it is a

'Binary classification'.

So, I am solving a dataset of a
'binary classification' problem.

Let's say,
over here, I have a ~~1,000 data~~
dataset 1,000 datapoints.

$\begin{array}{l} \rightarrow 900 - \text{yes, so } \boxed{1} \\ \rightarrow 100 - \text{No, so } \boxed{0} \end{array} \left. \vphantom{\begin{array}{l} \rightarrow 900 \\ \rightarrow 100 \end{array}} \right\} \text{Imbalanced dataset}$

Here, I have a 1,000 datapoints, this particular dataset say, 900 times, it is yes, which is $\boxed{1}$
100 times, it say ~~No~~ $\boxed{0}$, which is $\boxed{0}$

- * The output of the 900 datapoints is $\boxed{1}$
- * The output of the 100 datapoints is $\boxed{0}$.

Can we say, this is an 'imbalanced dataset'?

* Dumb Model $\rightarrow 1 = 90\%$ Accuracy = ~~Sufficient~~

If I create a dumb model, if I say a model just give output as $\boxed{1}$.

If I try to apply, this dataset,

* what accuracy we will get?

Will we get 90% accuracy of a dumb model.

This model only gives $\boxed{1}$ output

So, will this model gives a 90% accuracy?

Is this good or bad?

If we only dependent on Accuracy, will not be sufficient. ~~X~~

Because, if imbalance dataset, I am still getting 90% Accuracy.

But, my model just giving 1 kind of output.

In this scenario, we can't just dependent on accuracy.

So, we need to focus on 'precision' recall & 'F-Beta score'.

* (3) precision :-

Formula :-

$$\frac{TP}{TP + FP}$$

		(y) Actual value	
		1	0
(y') Predicted value	1	TP	FP
	0	FN	TN

Over here 'precision', what we are focusing on.

Can we write based on this equation.

* So, out of all the actual values, how many are correctly predicted.

Actual results

Actual value (Y)

1	0
1	0
TP	FP
FN	TN

predicted value (y)

So, here, I am actually getting with respect to this.

$$\frac{TP}{TP + FP}$$

This scenario, basically called as 'precision'.

In 'precision', we try to focus on (FP)

False/positive.

our main aim is to reduce False positive.

* problem statement of Question 1

In which problem statement, False/positive is very important.

Let's say, Question 1
my problem statement is to predict whether a 'male' is a ~~spam~~ spam or Ham?

Now, let's solve this problem statement,

why this False/positive statement is important.

① If male is a spam, ⁽¹⁾
it predicted. the spam ⁽¹⁾

		actual	
		1	0
predicted	1	TP	FP
	0	FN	TN

Then it is good [1 & 1] the model is

doing the right prediction.

② If a male is a spam (1) &

But, if it is predicted as a

ham (0); that is (FN) False Negative.

		actual	
		1	0
predicted	1	TP	FP
	0	FN	TN

It is predicted as a ~~spam~~ 'ham', (0)
then we will received the 'male' (1).

Yes, we will received the 'male'.

Does this ~~harm~~ harm us.

No right we just check it ~~about~~
of spam. This is fine.
we don't focus on ~~FP~~ (FN)

False/Negative. Reducing the (FN).

One thing note that, (FN) is
also an Error.

(FP) False/positive also an Error.

③ Now, see in the other scenario,

If a 'male' is not a spam,

If the male is 'Ham', but
my model is predicted
as a spam.

	1	0
1	TP	FP
0	FN	TN

Is this a critical

situation or not?

May I miss important emails.

* we received alot of emails, on daily basis,
~~class~~ today, class emails, etc....
live

what we are missing?

we are missing critical classes.

This is a ~~criti~~ critical problem, so this scenario,
Should I focus on (FP) False positive or
~~False~~ (FN) False Negative.

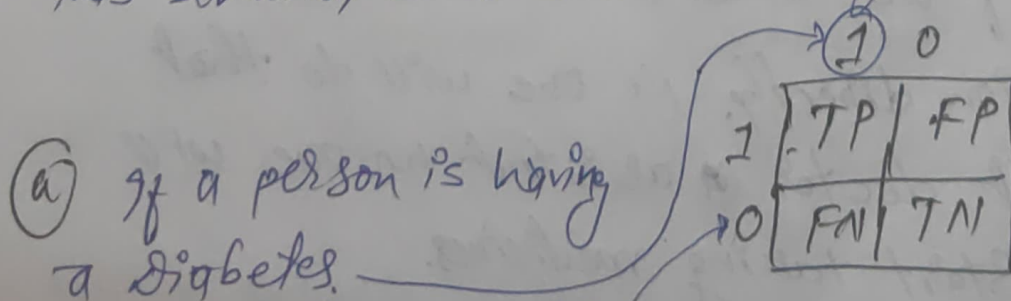
This works even in 'imbalance dataset'.

* I have to focus on reducing (FP) False positive.

* Question: 2 :-

Let's say, I have created an model,
whether a person is having ~~the~~ 'Diabetes' or
not diabetes? or Covid or any diseases?

This scenario, what we have to focus on



But my model predicted, it ~~is~~ does
not have diabetes.
Then he will not have medicine.

Then after sometime, he will die.

what we focus on reducing?

• we have to focus on

(FN) False Negative

	1	0
1	TP	FP
0	FN	TN

⑤ Let's go with the (FP) False positive

• let's say, A person is not having ~~die~~ diabetes, but my model says, you are having diabetes.

Then, he will go to the Hospital & check Health.

Then later he will get the result.

Going Hospital & giving his blood is fine, if we missed check up. we missed to check up with Doctor. what will happen.

what if we directly take ~~medicine~~ medication directly. No one will do that, without doctor's assistant no one will directly start taking medicines.

* So, what we focus on in case of diseases. we should focus on (FN) False Negative.

* So, for this, we are also using another metrics, called 'Recall'.

(4) Recall

Formula:

$$\frac{TP}{TP + FN}$$

definition: out of all the predicted values, how many are correctly predicted.

Let's take a scenario,

Tomorrow the stock market is going to crash.

If this is our scenario, what we have to focus on, whether (FN) or (FP) which one?



→ Consumer - FN (False Negative)
→ Companies - FP (False Positive)

* In this scenario, there is 2 points

- ① Consumer
- ② Companies

* we have to see, for whom we are creating this model.

Is it for consumer or is it for Companies?

* If we are creating for 'Consumer', what is most important thing.

we probably have to reduce the (FN) False Negative.

* If we are creating for the 'Companies', we should try to reduce the ~~False Negative (FN)~~ False positive (FP).

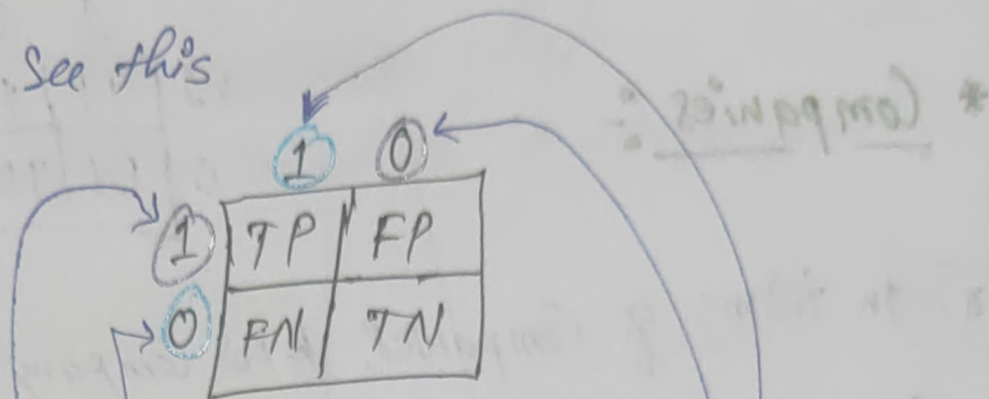
If the 'Companies' come to know this right they will take the secondary action.

* If the 'Consumers' come to know ~~they~~ with respect to (FN) False Negative, they ~~will~~ need to reduce ~~it~~ it.

So, that we will be able to do the prediction.

why?

Let's see this



* Consumers :-

- (a) Now, with respect to the consumers, what happens, tomorrow the stock market, going to fail. But model predict, that it is not going to fail.

what will happen, consumer's money gone, investors from outside will take money & they will go, huge loss will happen.

- (b) Let's say, tomorrow, over here we can see that, tomorrow it is saying that market is not going to plunge (good)

But it is predicting that, market will plunge.

This scenario, it will not impact that the customers, consumer will just sell & go, No Loss.

~~But in this~~

* Companies :-

	1	0
1	TP	FP
0	FN	TN

(a) In terms of Companies, For Company, it says that the stock market is not going to crash.
↓
tomorrow

But, it is crashed.

In this case (FP) False positive means to be reduce.

So, that Companies can take a ~~pro~~ 'appropriate' action, ~~provide some good offers~~, like

(i) provide some good offers.

(ii) ~~share~~ sell the share at discounted price.

all those steps they can take.

(b) In this case ~~base on~~ Some time based on the problem statement our (FP) False positive can also important & (FN) False Negative. can also been important.

~~(5)~~ In this scenario, what we do, we used something called as 'F-Beta Score'.

(5) F-Beta score

Formula :-

$$F_\beta = \frac{(1 + \beta^2) \times (\text{precision} \times \text{Recall})}{(\beta^2 \times \text{precision} + \text{Recall})}$$

Beta square

Here, we are using both ~~prec~~ precision & Recall.

① 1st Scenario :-

If (FP) False positive & (FN) False negative are both important.
Then β or Beta value will be 1.

$$\boxed{\beta = 1}$$

If it is $\beta = 1$, then, it will become,

$$\boxed{F1 \text{ Score}}$$

How we can write ~~(F1 Score)~~

$$F1 \text{ score} = 2 \times \frac{P \times R}{P + R}$$

$P = \text{precision}$
 $R = \text{Recall}$

② 2nd scenario :

If (FP) False positive is more important than (FN) False Negative.

Then the β or Beta value become $\boxed{0.5}$

$$\boxed{\beta = 0.5}$$

So, it becomes over here as

~~$$F_{0.5} \text{ score} = \frac{(1 + 0.25) \times \frac{P \times R}{P + R}}{(0.25 \times P) + R}$$~~

$$F_{0.5} \text{ score} = (1 + 0.25) \times \frac{P \times R}{(0.25 \times P) + R}$$

$$\boxed{F_{0.5} \text{ score} = 1.25 \times \frac{P \times R}{0.25P + R}}$$

③ 3rd scenario :

If FN (False Negative) is important than False positive (FP). then,

we write,

$$F2 \text{ Score} = \frac{(1 + 2^2) * P * R}{(2^2 * P) + R}$$

$$F2 \text{ Score} = \frac{(1 + 4) * P * R}{(4 * P) + R}$$

$$F2 \text{ Score} = \frac{.5 * P * R}{4P + R}$$

* So, this is with respect to the entire performance metrics.

*Note:-

(1) If we have a 'embalance dataset', we have to find out that, our (FP) is important, then we have to focus more into the precision (P).

(2) If we have a 'embalance dataset', if there we find out that our (FP) False positive is important, then we have focus more into precision (P).

③ If we have 'Emblance dataset' where says. False Negative (FN) is important then, we have to focus on Recall. & we have to reduce False Negative (FN).

* Interview Question

They will give a problem statement & tell you to focus on whether it ~~false~~ (FP) False positive or False Negative (FN).

practical. Continue. —

Started : 2:15:00 By (Sunny)

Logistic Regression

* dataset : Diabetes